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### Heat dissipation structure and electronic device using same

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#### Abstract

An electronic device includes a heat dissipation structure including a first heat dissipation member and a second heat dissipation member. The first heat dissipation member is disposed on a circuit board and is made of a thermally conductive material. The second heat dissipation member is disposed close to an antenna. A projection of the second heat dissipation member on an output surface of the antenna at least covers a portion of the output surface of the antenna. The second heat dissipation member is in contact with the first heat dissipation member. The second heat dissipation member is made of a plastic material mixed with a plurality of ceramic particles.

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## Background/Summary

### BACKGROUND OF THE INVENTION

#### Technical Field

(1) The present invention relates generally to a heat dissipation structure, and more particularly to a heat dissipation structure suitable for an antenna without interfering a signal of the antenna.

#### Description of Related Art

(2) It is known that various electronic devices are becoming more and more popular. When the operating speed or the signal transmission speed is getting faster and faster, a lot of heat is usually generated inside a casing of an electronic device. If the aforementioned heat is not effectively removed, the operating temperature of the electronic device will increase, thereby affecting the working efficiency of the electronic device or reducing the service life of the electronic device

(3) For example, in a wireless router having an antenna inside the wireless router, a metal component is generally connected to a heat source inside the wireless router to guide heat of the heat source out of the wireless router, so that the purpose of heat dissipation can be achieved. However, since metal will attenuate or shield a signal outputted by the antenna, the metal component cannot be disposed within a range of the antenna. As a result, the heat dissipation efficiency is limited, resulting in the problem that the working temperature in the wireless router is too high.

(4) Therefore, in order to achieve the purpose of improving the heat dissipation efficiency, a housing of the wireless router is generally enlarged to increase a heat dissipation space or a fan is generally disposed in the wireless router. However, such heat dissipation ways will increase the overall size of the wireless router or increase the cost. Therefore, how to provide a heat dissipation structure, which could improve the heat dissipation efficiency without interfering the signal of the antenna, has become a major issue in the industry.

#### BRIEF SUMMARY OF THE INVENTION

(5) In view of the above, the primary objective of the present invention is to provide a heat dissipation structure, which could improve the heat dissipation efficiency without interfering a signal of an antenna.

(6) The present invention provides a heat dissipation structure disposed in an electronic device. The electronic device includes a circuit board and an antenna disposed on the circuit board. The heat dissipation structure includes a first heat dissipation member and a second heat dissipation member. The first heat dissipation member is disposed on the circuit board. The first heat dissipation member is made of a thermally conductive material. The second heat dissipation member is disposed close to the antenna. A projection of the second heat dissipation member on an output surface of the antenna at least covers a portion of the output surface of the antenna. The second heat dissipation member is in contact with the first heat dissipation member. The second heat dissipation member is made of a plastic material, wherein the plastic material is mixed with a plurality of ceramic particles.

(7) In an embodiment, the ceramic particles include boron nitride. An addition of boron nitride accounts for 30-80 wt % of the second heat dissipation member.

(8) In an embodiment, the plastic material is thermoplastic or thermoset plastic.

(9) In an embodiment, a dielectric constant of the second heat dissipation member ranges between 3.5 and 4.4.

(10) In an embodiment, a thermal conductivity coefficient of the second heat dissipation member ranges between 7.5 W/mK and 15.5 W/mK.

(11) In an embodiment, the second heat dissipation member includes a first portion and a second portion connected to the first portion. A projection of the first portion on the output surface of the antenna at least covers the portion of the output surface of the antenna. The second portion has a first surface and a second surface opposite to the first surface. The first surface faces the circuit board. The second surface is tightly attached to the first heat dissipation member.

(12) In an embodiment, the first heat dissipation member includes a bottom heat sink, a heat dissipation unit, and a top heat sink. The heat dissipation unit is disposed between the top heat sink and the bottom heat sink. A bottom of the bottom heat sink is in contact with a heat-generating member on the circuit board. A side of the top heat sink facing the circuit board is tightly attached to the second surface of the second heat dissipation member.

(13) In an embodiment, a surface of the antenna opposite to the output surface faces the circuit board. A space is formed between the antenna and the circuit board. A portion of the bottom heat sink is disposed in the space.

(14) The present invention further provides an electronic device, including the aforementioned heat dissipation structure, a housing, an antenna, and a circuit board. The housing has a receiving space, wherein the heat dissipation structure is received in the receiving space. An inner wall of the

housing is connected to the first heat dissipation member. The antenna is disposed in the receiving space. The circuit board is disposed in the receiving space, wherein the antenna is disposed on the circuit board.

(15) In an embodiment, the inner wall of the housing is connected to the second heat dissipation member.

(16) With the aforementioned design, by the second heat dissipation member being made of the plastic material and the plastic material being mixed with the ceramic particles, the second heat dissipation member could simultaneously have a high thermal conductivity coefficient and a low dielectric constant, thereby improving the heat dissipation efficiency of the electronic device without interfering with the signal outputted by the antenna.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) The present invention will be best understood by referring to the following detailed description of some illustrative embodiments in conjunction with the accompanying drawings, in which
- (2) FIG. 1 is a perspective view of the heat dissipation structure according to an embodiment of the present invention;
- (3) FIG. 2 is a schematic view, showing a part of the heat dissipation structure according to the embodiment of the present invention is exploded;
- (4) FIG. 3 is a side view of the heat dissipation structure according to the embodiment of the present invention;
- (5) FIG. 4 is a side view of the heat dissipation structure according to the embodiment of the present invention seen from another direction;
- (6) FIG. 5 is a schematic view showing the temperature simulation result of the electronic device according to the comparative example 1 of the present invention and the temperature simulation results of the electronic device according to the embodiment 1 and the embodiment 2 of the present invention;
- (7) FIG. 6 is a schematic view showing the simulation result of the antenna scattering parameter of the electronic device according to the embodiment 1 of the present invention;
- (8) FIG. 7 is a schematic view showing the simulation result of the antenna scattering parameter of the electronic device according to the embodiment 2 of the present invention;
- (9) FIG. 8 is a schematic view showing the simulation result of the antenna scattering parameter of the electronic device according to the comparative example 1 of the present invention;
- (10) FIG. 9 is a schematic view showing the simulation result of the antenna scattering parameter of the electronic device according to the comparative example 1 of the present invention and the simulation results of the antenna scattering parameters of the electronic device according to the embodiment 1 and embodiment 2 of the present invention;
- (11) FIG. 10 is a schematic view showing the simulation result of the radiation pattern of the electronic device according to the embodiment 1 of the present invention;
- (12) FIG. 11 is a schematic view showing the simulation result of the radiation pattern of the electronic device according to the embodiment 2 of the present invention;
- (13) FIG. 12 is a schematic view showing the simulation result of the radiation pattern of the electronic device according to the comparative example 1 of the present invention; and
- (14) FIG. 13 is a schematic view showing the simulation result of the radiation pattern of the electronic device according to the comparative example 1 of the present invention and the simulation results of the radiation pattern of the electronic device according to the embodiment 1 and embodiment 2 of the present invention.

### DETAILED DESCRIPTION OF THE INVENTION

(15) A heat dissipation structure according to an embodiment of the present invention is illustrated in FIG. 1 to FIG. 4. As shown in FIG. 4, an electronic device **1** includes a housing **20**, two antennas **30**, the heat dissipation structure, and a circuit board **40**. The housing **20** has a receiving space **S** therein. The heat dissipation structure, the antennas **30**, and the circuit board **40** are received in the receiving space **S**. The antennas **30** are disposed on the circuit board **40** and are electrically connected to the circuit board **40**. The circuit board **40** is a printed circuit board (PCB). In the current embodiment, the electronic device **1** is a wireless router as an example, and each of the antennas **30** is a planar inverted-F antenna (PIFA) as an example. In other embodiments, the electronic device **1** could be another type of electronic devices, and the antennas **30** could be another type of antennas. In the current embodiment, the number of the antennas **30** is two as an example. In other embodiments, the number of the antenna **30** could be one or more.

(16) In the current embodiment, the housing **20** is approximately in a cuboid shape. In practice, the housing **20** could be a housing in different shapes (for example: a cylindrical housing). As shown in FIG. 4, the circuit board **40** is disposed at a position close to a bottom of the housing **20**. In the current embodiment, the two antennas **30** are disposed on the circuit board **40** and are respectively disposed at a position close to a left side of the circuit board **40** and a position close to a right side of the circuit board **40**. An electronic component **42** is disposed on the circuit board **40** and is located between the two antennas **30**.

(17) As shown in FIG. 1 and FIG. 2, the heat dissipation structure includes a first heat dissipation member **12** and a second heat dissipation member **14**. As shown in FIG. 3, the first heat dissipation member **12** is disposed on the circuit board **40**. The first heat dissipation member **12** is made of a thermally conductive material. For example, the thermally conductive material of the first heat dissipation member **12** could be a metal material (such as silver, copper, or aluminum), carbon fibers, a composite material, or any materials with a thermal conductivity coefficient greater than 90 W/mK, but not limited thereto.

(18) The first heat dissipation member **12** includes a bottom heat sink **121**, a heat dissipation unit, and a top heat sink **123**. The top heat sink **123** is disposed above the bottom heat sink **121** and is substantially parallel to the bottom heat sink **121**. The heat dissipation unit is a plurality of heat dissipation fins **122**. The heat dissipation fins **122** are disposed between the top heat sink **123** and the bottom heat sink **121**, wherein two opposite ends of each of the heat dissipation fins **122** are respectively connected to the top heat sink **123** and the bottom heat sink **121**. The heat dissipation fins **122** are arranged at constant intervals in a horizontal direction and are substantially parallel to one another. In the current embodiment, the electronic component **42** is a heat-generating member **42**. A bottom of the bottom heat sink **121** is in contact with the heat-generating member **42** on the circuit board **40**, so that a heat energy generated by the heat-generating member **42** on the circuit board **40** could be guided by the first heat dissipation member **12** to be discharged upward. In addition, a thermal pad with a thickness of 1~1.5 mm could be disposed on the circuit board **40** and the bottom heat sink **121**, so that the heat dissipation efficiency could be increased. In the current embodiment, the heat dissipation unit is a plurality of heat dissipation fins **122** as an example. In other embodiments, the heat dissipation unit could be any structures that could transfer heat energy; for example, the heat dissipation unit could be a heat sink in a C-shape or an aluminum block disposed between the top heat sink **123** and the bottom heat sink **121** that could also achieve the effect of heat transfer and heat dissipation, but not limited thereto.

(19) As shown in FIG. 2, the second heat dissipation member **14** is in a plate shape and is made of a plastic material. The plastic material is mixed with a plurality of ceramic particles, wherein the ceramic particles include boron nitride. An addition of boron nitride accounts for 30-80 wt % of the second heat dissipation member **14**; when the addition of boron nitride is greater than or equal to 30 wt % of the second heat dissipation member **14**, the second heat dissipation member **14** could have a higher thermal conductivity coefficient; when the addition of boron nitride is less than or equal to 80 wt % of the second heat dissipation member **14**, the second heat dissipation member **14**

would not cause too much interference to a signal outputted by the antennas **30**, and the molding effect of the plastic material would not be affected. The plastic material could be a thermoplastic or a thermoset plastic. By the plastic material being mixed with the ceramic particles, the second heat dissipation member **14** could have a low dielectric constant and the high thermal conductivity coefficient, wherein the dielectric constant of the second heat dissipation member **14** ranges between 3.5 and 4.4, and the thermal conductivity coefficient of the second heat dissipation member **14** ranges between 7.5 W/mK and 15.5 W/mK. In this way, the second heat dissipation member **14** could provide the function of heat conduction without interfering the signal outputted by the antennas **30**.

(20) As shown in FIG. 2, the second heat dissipation member **14** is disposed close to the antennas **30**. A projection of the second heat dissipation member **14** on an output surface **32** of the antennas **30** at least covers a portion of the output surface **32** of the antennas **30**. The second heat dissipation member **14** is in contact with the first heat dissipation member **12**. The second heat dissipation member **14** includes a first portion **141** and a second portion **142** connected to the first portion **141**. A projection of the first portion **141** on the output surface **32** of the antennas **30** at least covers the portion of the output surface **32** of the antennas **30**. The second portion **142** has a first surface **142a** and a second surface **142b** opposite to the first surface **142a**. The first surface **142a** faces the circuit board **40**. The second surface **142b** faces the first heat dissipation member **12** and is tightly attached to a side of the top heat sink **123** of the first heat dissipation member **12** facing the circuit board **40**; preferably, referring to FIG. 3, a left side and a right side of the top heat sink **123** are also in contact with the second heat dissipation member **14**; a surface of the antennas **30** opposite to the output surface **32** faces the circuit board **40**, and a space is formed between the antennas **30** and the circuit board **40**; a portion of the bottom heat sink **121** is disposed in the space.

(21) In this way, the heat energy generated by the heat-generating member **42** on the circuit board **40** could not only be guided by the first heat dissipation member **12** to be discharged upward, but could also be guided to the second heat dissipation member **14** by the first heat dissipation member **12** to be discharged outward. Compared with the conventional ways of improving the cooling efficiency by enlarging a housing to increase a heat dissipation space or by disposing a fan, the second heat dissipation member **14** is disposed in the space above the output surface **32** of the antennas **30**, so that the heat dissipation structure of the present invention could achieve the purpose of improving heat dissipation efficiency without increasing the volume of the electronic device **1**.

(22) As shown in FIG. 4, an inner wall of the housing **20** is connected to the first heat dissipation member **12** and the second heat dissipation member **14**. For example, the inner wall of the housing **20** could be connected to the first heat dissipation member **12** and the second heat dissipation member **14** through a thermally conductive double-sided tape, a thermal pad, or a thermal grease, but not limited thereto.

(23) Two embodiments 1-2 and a comparative example 1 of the present invention are illustrated in the below description. As shown in Table 1, an electronic device of the comparative example 1 has the first heat dissipation member **12** but not the second heat dissipation member **14**, while both an electronic device of the embodiment 1 and an electronic device of the embodiment 2 have the first heat dissipation member **12** and the second heat dissipation member **14**. Referring to simulation results as shown in FIG. 5, it can be seen that a temperature of the electronic device near a central region of the electronic device in the comparative example 1 is 81.91 degrees Celsius during operation, while a temperature of the electronic device near a central region of the electronic device in the embodiment 1 is 79.54 degrees Celsius during operation, and a temperature of the electronic device near a central region of the electronic device in the embodiment 2 is 79.35 degrees Celsius during operation. Compared with the comparative example 1, the temperature of the electronic device of the embodiment 1 could be reduced by 2.37 degrees Celsius, and the temperature of the electronic device of the embodiment 2 could be reduced by 2.56 degrees Celsius. In this way, it can

be clearly seen that the second heat dissipation member **14** could improve the heat dissipation efficiency and thus reduce the temperature of the electronic device **1**. In addition, the performance of heat dissipation of the embodiment 1 and the performance of heat dissipation the embodiment 2 show that when the addition of boron nitride is greater, the second heat dissipation member **14** could have a higher thermal conductivity coefficient so that the heat dissipation efficiency could be improved.

(24) Furthermore, as shown in Table 1 and FIG. **6** to FIG. **9**, an antenna return loss of the comparative example 1 at 2.05 GHz is  $-18.47$  dB, an antenna return loss of the embodiment 1 at 2.05 GHz is  $-14.87$  dB, and an antenna return loss of embodiment 2 at 2.05 GHz is  $-13.5$  dB. A loss percentage of the antenna return loss of the comparative example 1 to the antenna return loss of the embodiment 1 is 19.5%, and a loss percentage of the antenna return loss of the comparative example 1 to the antenna return loss of the embodiment 2 is 26.9%. In other words, the influence of the second heat dissipation member **14** on a S-Parameter of the antennas **30** is limited. In addition, as shown in FIG. **10** to FIG. **13**, an antenna radiation value of the embodiment 2 is 0.65 dB lower than an antenna radiation value of the comparative example 1. In other words, the influence of the second heat dissipation member **14** on a radiation pattern of the antennas **30** is limited.

(25) TABLE-US-00001 TABLE 1 Second Addition Thermal Antenna S- heat of boron conductivity Temperature parameter at dissipation nitride coefficient Dielectric reduced 2.05 GHz member (wt %) (W/mK) constant ( $^{\circ}$  C.) (dB) Comparative No — — — —  $-18.47$  Example 1 Embodiment 1 Yes 60 8 4.1 2.37  $-14.87$  Embodiment 2 Yes 75 15 4.4 2.56  $-13.5$

(26) With the aforementioned design, by the second heat dissipation member **14** being made of the plastic material and the plastic material being mixed with the ceramic particles, the second heat dissipation member **14** could simultaneously have a high thermal conductivity coefficient and a low dielectric constant, thereby improving the heat dissipation efficiency of the electronic device **1** without interfering with the signal outputted by the antennas **30**.

(27) It must be pointed out that the embodiments described above are only some preferred embodiments of the present invention. All equivalent structures which employ the concepts disclosed in this specification and the appended claims should fall within the scope of the present invention.

## Claims

1. A heat dissipation structure, disposed in an electronic device; the electronic device comprises a circuit board and an antenna disposed on the circuit board; the heat dissipation structure comprising: a first heat dissipation member disposed on the circuit board, wherein the first heat dissipation member is made of a thermally conductive material; and a second heat dissipation member disposed close to the antenna, wherein a projection of the second heat dissipation member on an output surface of the antenna at least covers a portion of the output surface of the antenna; the second heat dissipation member is in contact with the first heat dissipation member; the second heat dissipation member is made of a plastic material, wherein the plastic material is mixed with a plurality of ceramic particles, wherein the second heat dissipation member comprises a first portion and a second portion connected to the first portion; a projection of the first portion on the output surface of the antenna at least covers the portion of the output surface of the antenna; the second portion has a first surface and a second surface opposite to the first surface, wherein the first surface faces the circuit board, and the second surface is tightly attached to the first heat dissipation member.

2. The heat dissipation structure as claimed in claim 1, wherein the plurality of ceramic particles comprise boron nitride; an addition of boron nitride accounts for 30-80 wt % of the second heat dissipation member.

3. The heat dissipation structure as claimed in claim 1, wherein the plastic material is thermoplastic

or thermoset plastic.

4. The heat dissipation structure as claimed in claim 1, wherein a dielectric constant of the second heat dissipation member ranges between 3.5 and 4.4.
  5. The heat dissipation structure as claimed in claim 1, wherein a thermal conductivity coefficient of the second heat dissipation member ranges between 7.5 W/mK and 15.5 W/mK.
  6. The heat dissipation structure as claimed in claim 1, wherein the first heat dissipation member comprises a bottom heat sink, a heat dissipation unit, and a top heat sink; the heat dissipation unit is disposed between the top heat sink and the bottom heat sink; a bottom of the bottom heat sink is in contact with a heat-generating member on the circuit board; a side of the top heat sink facing the circuit board is tightly attached to the second surface of the second heat dissipation member.
  7. The heat dissipation structure as claimed in claim 6, wherein a surface of the antenna opposite to the output surface faces the circuit board; a space is formed between the antenna and the circuit board; a portion of the bottom heat sink is disposed in the space.
  8. An electronic device, comprising: the heat dissipation structure as claimed in claim 1; a housing having a receiving space, wherein the heat dissipation structure is received in the receiving space; an inner wall of the housing is connected to the first heat dissipation member; the antenna disposed in the receiving space; and the circuit board disposed in the receiving space, wherein the antenna is disposed on the circuit board.
  9. The electronic device as claimed in claim 8, wherein the inner wall of the housing is connected to the second heat dissipation member.
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